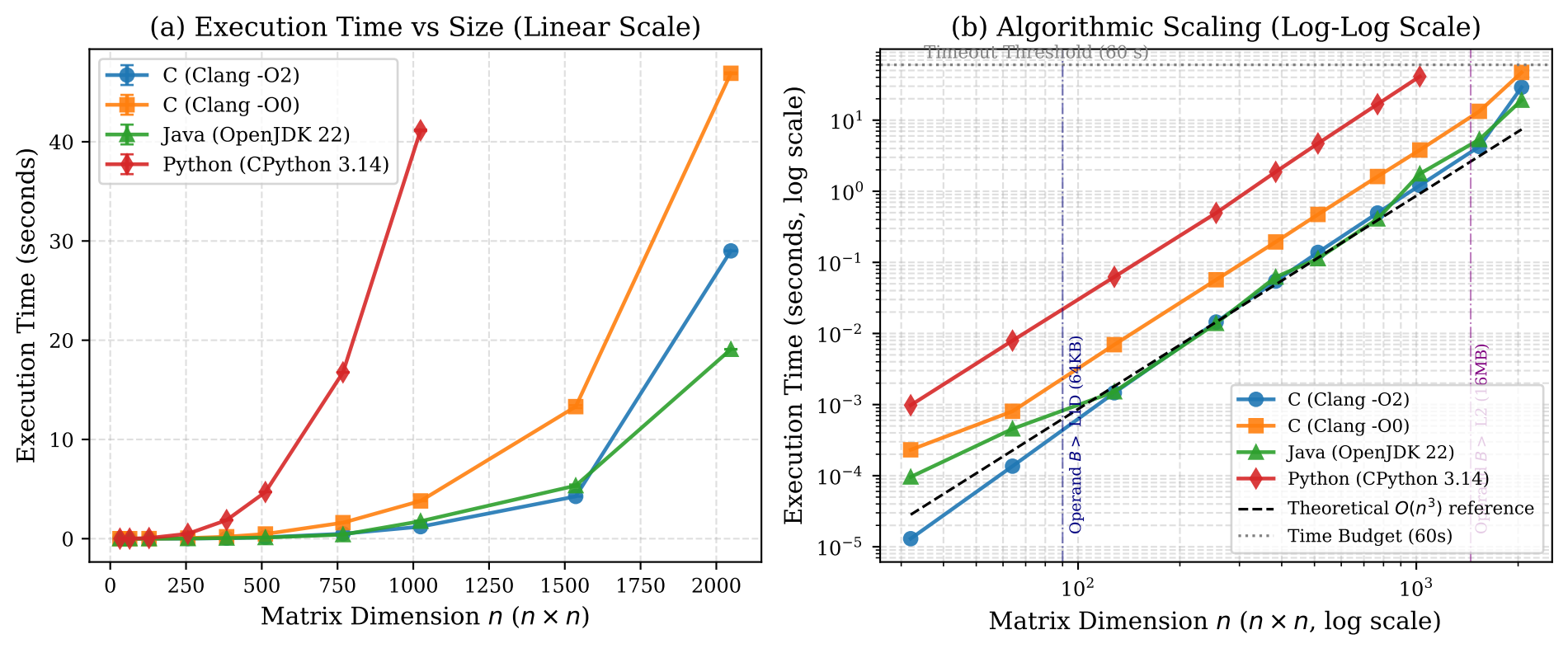
(a) Execution Time vs Size (Linear Scale)
C (Clang -O2)
C (Clang -O0)
Java (OpenJDK 22)
Python (CPython 3.14)
Execution Time (seconds)
Matrix Dimension n ($n \times n$)
(b) Algorithmic Scaling (Log-Log Scale)
Timeout Threshold (60 s)
Operand B > L1D (64KB)
Operand B > L2 (16MB)
Theoretical $O(n^3)$ reference
Time Budget (60s)
Execution Time (seconds, log scale)
Matrix Dimension n ($n \times n$, log scale)